# Layout (≈*1,681*)

## Component Object

Component Name | Component Value | Placement Position | Placement Layer

**Component Example**

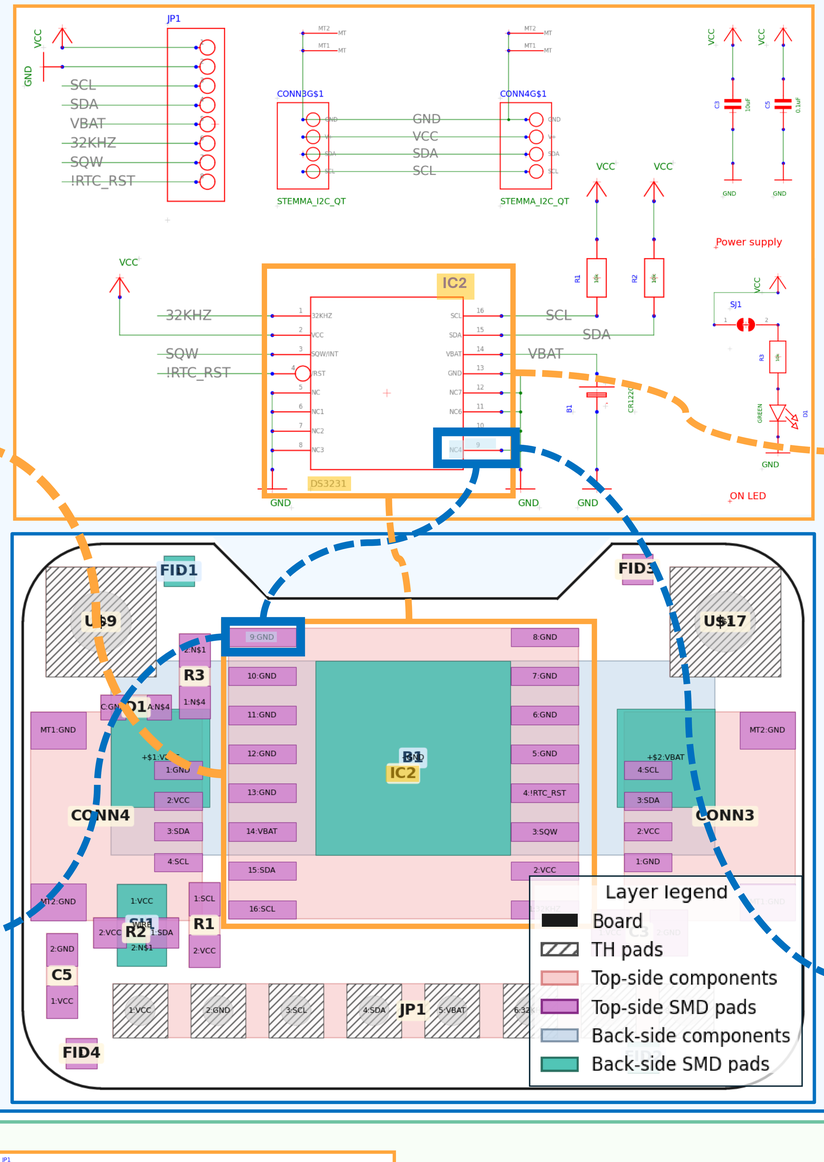
**Comp. Name:** IC2 **Comp. Value:** DS3231
**Comp. Type:** Real Time Clock IC **Comp. Orientation:** 90°
**Comp. Position:** [12.3825, 10.287] **Pad List:** [9, …]
**Comp. Placement Layer:** Top-side
**Comp. Overall Dimension:** 11.2 x 10.4 mm

## Soldering Pad Object

Pad Name | Pad Dimension | Placement Position | Component Name

**Solder Pad Example**

**Pad Name:** 9 **Net Name:** GND
**Pad Type:** Surface Mount **Pad Orientation:** 0°
**Pad Position:** [4.445, 4.6]
**Pad Placement Layer:** Top-side
**Pad Overall Dimension:** 0.6 x 2.2 mm




# Schematic (≈*1,681*)

## Symbol Object

Symbol Name | Symbol Value | Symbol BBox | Symbol Type

**Schematic Symbol Example**

**Symbol Name:** IC2 **Symbol Value:** DS3231
**Symbol Type:** Real Time Clock IC **Pin List:** [NC4, …]
**Symbol BBox:** [156, 168, 304, 304]
**Symbol Name Txt BBox:** [182, 293, 208, 300]
**Symbol Value Txt BBox:** [182, 293, 208, 300]

## Pin Object

Pin Name | Pad Name | Pin BBox | Symbol Name

**Schematic Pin Example**

**Pin Name:** NC4 **Pad Name:** 9
**Symbol Name:** U2
**Pin BBox:** [260, 243, 305, 263]
**Pin Name Txt BBox:** [248, 244, 254, 253]
**Pad Name Txt BBox:** [309, 244, 331, 253]

# Netlist Graph (≈ *55.8K*)

## Graph Object

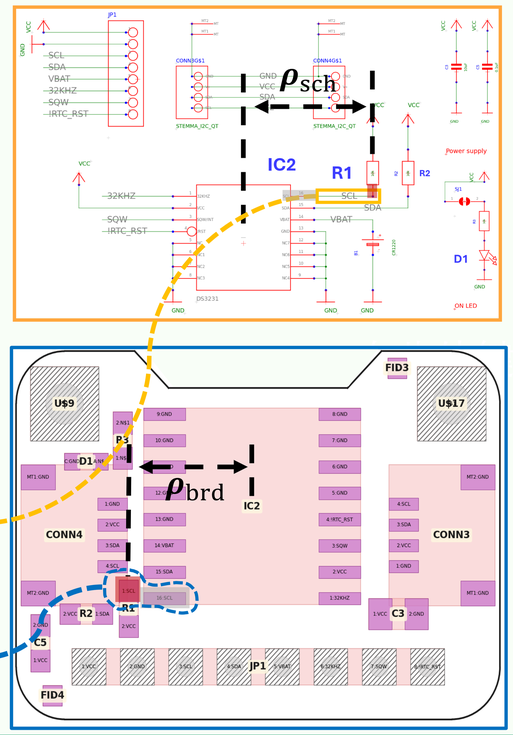
**Symbol Node (≈*99.66K*)** | **Comp. Node (≈*77.24K*)** | **Pin Node (≈*222.5K*)** | **Soldering Pad Node (≈*168.8K*)** | **Sch. Net (≈ *139.4K*)** | **Layout Net (≈*139.4K*)**

**Schematic Net Example**

**Net Name:** SCL **Spatial Weight :** 0.58 ($\rho_{sch}$)
**End Point Pin:** IC2_[16] **End Point Pin:** R1_[1]

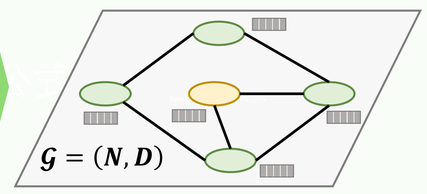
**Layout Net Example**

**Net Name:** SCL **Spatial Weight :** 0.51 ($\rho_{brd}$)
**End Point Pad:** IC2_[16] **End Point Pad:** R1_[1]

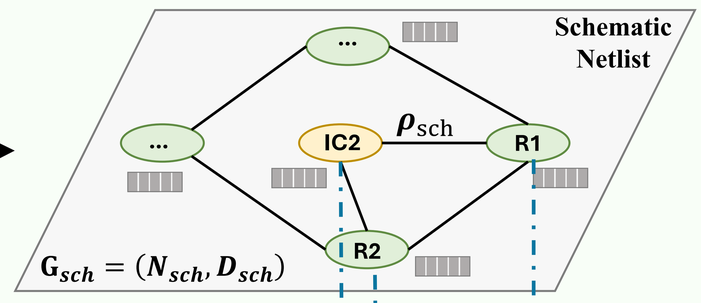


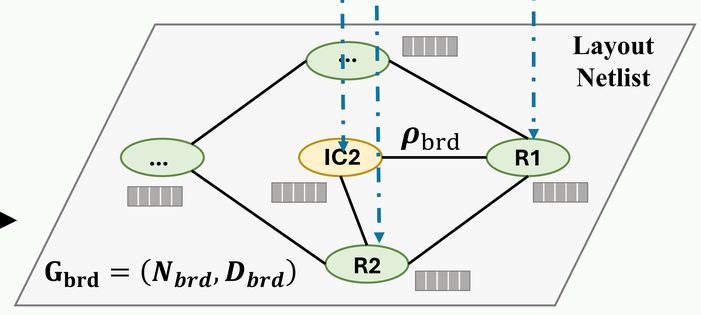

Encode **Visual Locality** onto **Schematic Coordinates**

Encode **Physical Locality** onto **Layout Coordinates**




**Netlist Graph**




# Printed-Circuit-Board Design (≈*1,681*)

## Meta Information

**Overview:**
The system is built around a Texas Instruments CC3220-series microcontroller that integrates Wi-Fi connectivity and analog sensing capabilities.

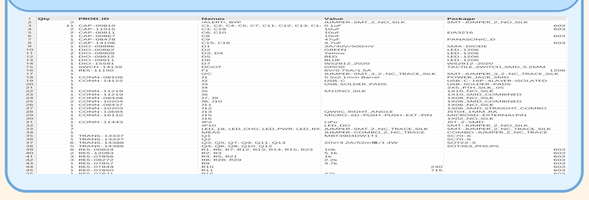
**Design Description:**

- **Microcontroller Core:** Texas Instruments CC3220-series MCU providing Wi-Fi connectivity, analog-to-digital conversion, and general-purpose digital interfaces.
- **Power Management:** Low-dropout voltage regulators and decoupling capacitors generate stable 3.3V
- …

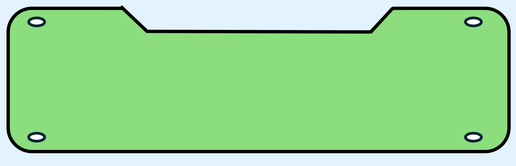
**Specifications:**

- Microcontroller: Texas Instruments CC3220 Wi-Fi MCU
- …

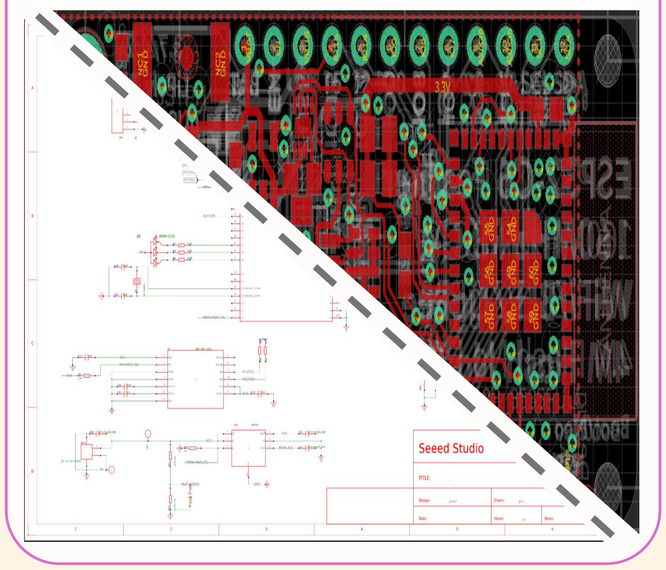
## BoM & Attachments



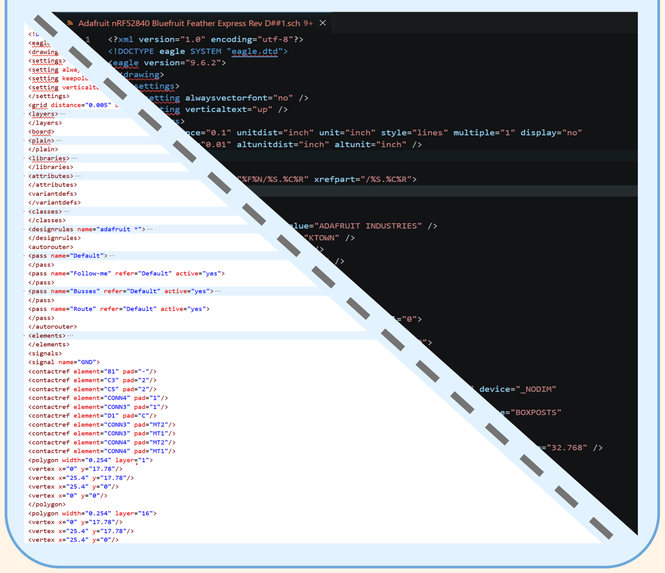
## Board Frame



## Sch. & Layout Diagram




## Source Code (Sch. & Layout)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